TOPSTECHNOLOGIES
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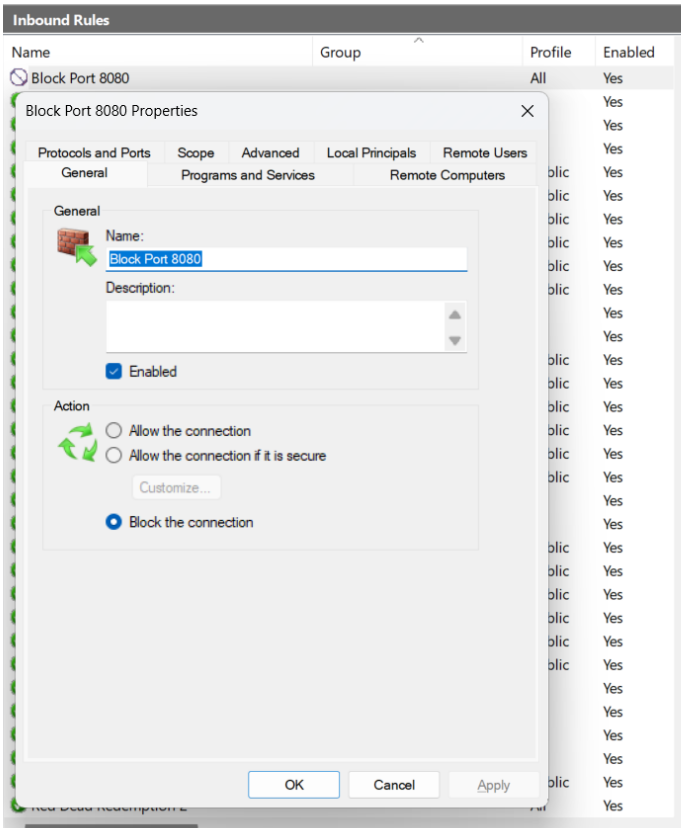
Secure networks with firewalls NAT, port filtering, packet filtering, and other methods

1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

ANS :

- Steps:
  1. Press Windows + R, type wf.msc, and press Enter.
  2. Click Inbound Rules → New Rule.
  3. Select Port → Next.
  4. Choose TCP and enter 8080 in Specific Local Ports.
  5. Click Next.
  6. Select Block the Connection.
  7. Apply to Domain, Private, Public.
  8. Name the rule: Block Port 8080
  9. Click Finish.



2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access. Hint: Use any drawing tool or even hand-draw and click a photo.

ANS :

- NAT Diagram

```
               INTERNET
           Public IP: 49.36.120.10
                    |
                    |
             +----------------+
             |  WiFi Router   |
             | NAT Enabled    |
             +----------------+
               /      |      \
              /       |       \
             /        |        \
            /         |         \
      Laptop        Phone       Smart TV
   192.168.1.2   192.168.1.3   192.168.1.4
```

- **Explanation**
  - The router assigns private IP addresses to devices inside the home network.
  - When devices access the internet, the router translates their private IPs into a single public IP using NAT.
  - This saves public IP addresses and adds a layer of security.

3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

ANS :

- **Configuration Steps**

1. Login to Router Admin Panel.
   - Example: http://192.168.1.1

2. Navigate to:
   - Security → Access Control
     or
   - Firewall → Port Filtering

3. Create Rule:

Setting	**Value**
Direction	Outbound
Protocol	TCP
Port	25
Action	Deny / Block

4. Save and Apply.

4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53. Hint: Focus on source/destination IPs and ports in the rule set.

ANS :

- Assumed Rule Set

Rule	**Action**
Allow HTTP (Port 80)	Allow
Allow DNS (Port 53)	Allow
Block SSH (Port 22)	Block

- Packet Evaluation

Packet	**Result**	**Reason**
HTTP request from 192.168.1.5 → 172.217.166.78:80	Allowed	HTTP Port 80 is allowed
SSH request from 10.0.0.10 → 34.201.23.1:22	Blocked	SSH Port 22 is blocked
DNS query from 192.168.1.7 → 8.8.8.8:53	Allowed	DNS Port 53 is allowed

- Answer

1. HTTP Request → **Allowed**
2. SSH Request → **Blocked**
3. DNS Query → **Allowed**

5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

ANS :

1. **Blocking Malicious Traffic**

- Firewalls detect and block suspicious requests such as DDoS attacks, brute-force login attempts, and unauthorized access attempts before they reach the servers.

2. **Restricting Access to Internal Services**

- Packet filtering ensures that only authorized systems can access databases, admin panels, and backend services, protecting sensitive user data.

3. **Filtering Application Requests**

- Firewalls inspect incoming traffic and block harmful payloads such as SQL Injection, Cross-Site Scripting (XSS), and bot-generated traffic to keep services secure and available.